AI Optimisation for Decarbonising Freight

Executive summary

Decarbonising road and maritime freight is a critical near-term challenge for the UK, with road freight generating 4.6% of UK emissions[1] and shipping 3%[2]. While decarbonising fuels (for example through electrification) remains the primary route to decarbonisation in the long term, improving operational efficiency is essential to reducing emissions. The Climate Change Committee's Carbon Budget 6[3] estimated efficiencies could deliver **a 9-11% reduction in HGV emissions** between 2019 and 2030.

A significant proportion of this could be achieved through smarter, AI-enabled optimisation of logistics. This Evidence Pack identifies three interlocking opportunities where optimisation can deliver meaningful emissions reductions to address the ADViCE Grand Challenge of Decarbonising Road and Freight[4]:

1. **What is transported, and where**: improving load consolidation, reducing empty running, and enabling aggregation across fleets.
2. **How vehicles move from A to B**: improving routing, journey management, and fuel efficiency for individual trips.
3. **How congestion is managed at endpoints**: reducing waiting, queuing, and idle time at depots, ports, warehouses, and charging locations.

Adoption of optimisation tools across these areas is mixed, but the general pattern is that the **binding constraints are organisational and systemic, rather than technical**. For example, AI tools optimising loads are primarily adopted by large operators, while small and medium-sized operators – who make up most trucks on the road – face structural barriers to AI adoption related to their fragmentation in the industry, limited data visibility, and weak coordination mechanisms between trucks of other fleets.

Maximising the benefits of AI to decarbonise freight therefore depends less on developing novel algorithms, and more on **enabling optimisation to work effectively across fragmented systems**.

Calls to action

- **Smarter targeting of AI R&D**, focusing on genuinely complex, system-level problems rather than isolated optimisation tasks.
- Address **system bottlenecks at endpoints,** such as standards, interfaces, or intermediaries that allow endpoint information to be shared across actors.
- Support **aggregation among smaller operators** (like pallet networks) to enable optimisation to scale for smaller organisations.

What counts as AI?

AI is an extremely broad term, which is often extended to cover existing approaches to optimisation. Whether an approach is strictly AI or not is relatively unimportant – what matters is whether it provides an effective and appropriate solution to a problem. However, understanding some of the different categories of AI optimisation tools can be helpful to quickly identify whether the tool being proposed is appropriate.

This is particularly important because Large Language Models (which are what people often mean when they say 'AI') are not the type of AI best suited to solve most optimisation problems. An outline of different types of optimisation can be found in the table below:

Type of optimisation	Basic approach	Where it is used
Heuristics, rules-based and spreadsheet - driven approaches (non-AI)	"Use fixed rules and human judgement based on experience"	By small and medium-sized fleets that still rely on manual or semi-automated approaches.
Exact optimisation methods/mathematical programming	"Create a mathematical model of each problem and find the provably correct solution"	The backbone of many commercial optimisation tools in use today - widely used for routing, scheduling, and load allocation. However, they struggle to scale to larger problems.
Evolutionary algorithms	"For each problem, iteratively try lots of solutions, combining good ones and discarding bad ones"	Often used to intelligently search for high-quality solutions to complex optimisation problems where exact mathematical methods are too slow or rigid.
Reinforcement learning and policy learning	"Learn effective decision strategies through feedback over time rather than solving each problem from scratch"	Explored for large-scale, highly dynamic problems in simulation and digital twin environments, but has yet to be widely used at scale.
Machine learning and deep learning	"Predict the near future from patterns in past data"	Indirectly supports optimisation by improving forecasts and estimates that feed into planning decisions, e.g. predicting demand, estimating travel or arrival times, and forecasting congestion or delays.

Opportunities to accelerate decarbonisation through optimisation

AI-enabled optimisation in freight breaks down into three distinct but interlocking opportunities, each operating at a different scale of the system and constrained by different limiting factors. These opportunities sit at the heart of whether AI is being applied to the right problems, at the right scale, and within systems capable of supporting it. The three areas are:

1. What is transported, and where
2. How vehicles move from A to B
3. How congestion is managed at endpoints

Optimisation of each of these stages can be effective in reducing emissions, regardless of whether fuels being used are high or low carbon. It is important to note that even when organisations optimise in these areas, they may be optimising for objectives other than emissions (time, revenue, customer satisfaction and cost). The optimal solution for reducing emissions might be suboptimal for consumer satisfaction or revenue in which case a company is unlikely to use it unless there are sufficient incentives to do so.

Optimisation opportunity 1: What is transported, and where

Load optimisation, demand aggregation, and empty run reduction

Load optimisation focuses on decisions about **how shipments are allocated across vehicles**, aiming to reduce empty running, improve vehicle fill rates, and allocate loads to minimise total distance travelled. Large fleet operators are proactively driving optimisation efforts here. For example, UPS delivers over 20 million packages daily with 125,000 vehicles using an AI routing system called ORION. In 2024, it reduced routes by 2-4 miles, saving 10 million gallons of fuel annually and cutting 100,000 metric tons of CO_2 emissions[5].

AI optimisation here aims to reduce total vehicle kilometres travelled by increasing load factors and reducing empty or partially loaded trips. This lowers fuel consumption and tailpipe emissions across the fleet. Industry and academic studies commonly cite ~10–30% reductions in distance travelled or fuel use[6] [7] when load optimisation (alongside route optimisation) is applied at sufficient scale, although real-world baselines are often unclear.

Adoption snapshot.
In practice, uptake of load optimisation varies sharply by organisation size. Large logistics operators typically use Transport Management Systems (e.g. Averion TMS, Descartes, Paragon, and Qargo) that incorporate exact or evolutionary algorithm optimisation methods to plan loads across their own fleets.

By contrast, most small and medium-sized operators (90% of road freight operators in the UK operate fleets of under 10 vehicles[8]) continue to rely on heuristics - manual planning, fixed rules, or spreadsheet-based tools, which do not scale effectively as complexity increases. Load optimisation for smaller fleets is much more challenging, with UK road freight statistics indicating that 30% of vehicle movements occur with empty or under-utilised loads[9], making this an important area to focus decarbonisation efforts.

Aggregation mechanisms such as pallet networks[10] represent a key mechanism for smaller operators to participate in load optimisation. These networks consolidate demand across many regional hauliers and have demonstrably improved vehicle utilisation within their hubs. Some pallet networks have begun introducing AI-labelled tools to improve sorting, forecasting, and operational planning (e.g. Palletforce's Alliance Sense[11] and Palletline's Palleteyes[12]).

Key constraint:

The benefits of this form of optimisation are very difficult to achieve at the level of individual vehicles or small fleets. So, for the many smaller operators in the UK participation in aggregation mechanisms – like pallet networks – is crucial for reducing fuel consumption and costs at a system-level. Improvements in cutting-edge optimisation algorithms are less likely to be important to those smaller operators, though they may still benefit larger operators.

Optimisation opportunity 2: How vehicles move from A to B

Route planning, speed optimisation, and journey management

Unlike load optimisation, which determines whether freight movements occur at all, route and journey optimisation focuses on improving the efficiency of trips that have already been planned. This includes optimising for route selection, speed profiles, sequencing of stops, and real-time rerouting in response to traffic, weather, or operational constraints. This is the most established area of optimisation in logistics and is often the first entry point for digital tools aimed at improving efficiency.

In practice, route and journey optimisation is delivered primarily using exact optimisation methods and evolutionary algorithms embedded within Transport Management Systems – similar to load optimisation. This aims to reduce fuel consumption per journey by shortening routes, avoiding congestion, smoothing driving patterns, and reducing idling. Machine learning models are commonly used to improve inputs such as travel time and congestion estimates[13]. Industry case studies and academic literature commonly report emissions reductions in the range of ~10–15%[14] from improved routing and journey management, with higher figures up to 30% produced through fleet management and dynamic route optimisation[15] using reinforcement learning. However, benefits are highly context dependent, and algorithmic improvements see marginal gains at best.

Autonomous vehicles, although not the focus of this evidence pack, may offer significant improvements in driving efficiency in the longer-term due to the ability to optimise driving behaviours more directly based on real-time data.

Adoption snapshot:
Compared with other optimisation opportunities, uptake of route optimisation tools is relatively widespread, particularly across large and medium-sized operators who use Transport Management Systems and navigation platforms that incorporate route planning, traffic awareness, and fuel optimisation. Smaller operators are also more likely to use some form of route optimisation than more complex load-aggregation tools, often relying on off-the-shelf navigation or basic digital planning tools. As a result, adoption gaps by organisation size are narrower here than in load optimisation.

Key constraint:
For most use cases, the algorithms themselves are already sufficiently capable. Traditional optimisation methods and established tools perform well for small and medium scale routing problems. The primary challenges are uneven adoption and limited system-level coordination. Importantly, route optimisation alone cannot address inefficiencies created elsewhere in the system (e.g. poor load allocation or destination bottlenecks) that increase emissions.

Optimisation opportunity 3: How congestion is managed at endpoints

Scheduling, queuing, and waiting time reduction

Endpoint congestion occurs when vessels and vehicles are forced to queue or idle at a destination, e.g. at ports, depots, warehouses, and increasingly charging locations for electrified freight. 10–25% of maritime emissions occur during activities like manoeuvring, anchoring, or berthing near the port[16]. Unlike route optimisation, which improves how journeys are executed, endpoint management focuses on whether vessels and vehicles can be serviced efficiently upon arrival and whether arrival times align with available capacity.

Endpoint management relies primarily on evolutionary algorithms and mathematical programming, but there is a rapidly growing number of machine learning approaches being developed – although largely in academic environments or at pilot stage[14 15]. There is growing interest in more dynamic, system-wide approaches, but evidence of large-scale deployment remains limited, and is hampered by challenges around sharing commercially sensitive data across participants[14].

This is important to address, as unproductive waiting and idling are already a major source of inefficiency in maritime logistics[17 18], where vessels frequently wait for extended periods, sometimes days, due to berth availability, tidal constraints, and scheduling mismatches. Studies of port and terminal scheduling systems suggest the potential for significant reductions in waiting times and associated emissions through the use of advanced optimisation approaches[15 19 20]. Examples of innovators addressing this problem include loadmaster.ai[21] and the Blue Visby Solution. The latter cites a ~15% reduction in emissions by eliminating “sail-fast-then-wait” behaviour through coordinated voyage planning and just-in-time arrivals[22]. The solution explicitly seeks to bridge both horizontal fragmentation (coordination across multiple vessels) and vertical fragmentation (misaligned incentives across owners, charterers, and ports)[23] optimising the system rather than vessels or an individual port[24].

Road freight may benefit from learning lessons from maritime here, as similar constraints are expected to emerge as electrification increases dwell times and tightens capacity at depots and charging infrastructure. Syslek’s ENROUTE system[25] is an example of what can be done in this space – monitoring vehicles and predicting arrival times to dynamically adjust charging schedules or route to alternative chargers.

Adoption snapshot:
Adoption of endpoint scheduling tools is uneven. Many ports and large logistics hubs operate appointment or booking systems to manage arrivals and departures, particularly in maritime logistics and at major distribution centres (e.g. Felixstowe, London Container Terminal, and DP Southampton). Like pallet networks, ports also participate in network services via five Community System Providers[26], which help coordinate customs clearance. Port booking systems typically focus on allocating time slots and managing capacity, rather than optimising across the wider transport system.

National Highways publishes extensive traffic data, but this is not freight-centric and focuses primarily on link-level conditions rather than congestion at fixed nodes such as depots, ports, warehouses, terminals, and charging locations. As a result, smaller depots and road freight destinations often have limited visibility of inbound arrivals and rely on heuristics - manual scheduling, fixed rules, or informal coordination.

Key constraint:
Optimising journeys without accounting for destination conditions can exacerbate delays, simply shifting congestion from roads to depots. Addressing this challenge requires endpoint aware optimisation, where routing, scheduling, and departure decisions account not just for travel time, but for real time conditions at the destination[27], such as yard capacity or charging occupancy.

This, in turn, depends on data availability, interoperability, and coordination across multiple actors, making endpoint management a system-level challenge. In the maritime sector, AI applications optimising endpoint congestion, like berth allocation and yard management, have been criticised for communication gaps between approaches, highlighting a risk the sector faces in developing novel applications in isolation[14]. This may be an inherent challenge for maritime and road freight compared to industries with more centralised traffic management systems like the aviation sector. Decentralised governance in these industries creates a complex operational landscape. Vessels follow global regulations set by the International Maritime Organization, while ports operate under national frameworks[28]. Effective solutions are therefore likely to benefit from standards, shared data frameworks, and coordination mechanisms, rather than AI interventions focused on individual vessels, vehicles, or operators.

Key recommendations for government

The current evidence suggests that accelerating the contribution of AI to freight decarbonisation will depend less on advancing AI research in isolation, and more on targeted prioritisation and the creation of enabling conditions that allow optimisation methods to operate effectively at scale.

1. Focus AI research & development on genuinely complex, system-level problems

Where advanced AI is likely to be most impactful is not in optimising individual vehicles or routes, but in problems that span multiple actors, assets, and decision layers. This includes co-optimisation across many fleets or operators, and optimisation spanning multiple areas such as routing, scheduling, and endpoints.

Public and collaborative R&D funding should hold a very high bar on applications of AI to optimisation of individual organisation operations and instead prioritise AI applications that:

- Address system-level barriers (e.g. aggregation, coordination, or coupling across the system).
- Demonstrate clear pathways to deployment at scale (not only simulation results).

This approach aligns investment with areas where traditional methods struggle to scale, and where approaches enabled by cutting-edge AI may unlock qualitatively new forms of optimisation.

2. Enable optimisation by improving data around bottlenecks at endpoints

Addressing endpoint congestion is not primarily an algorithmic challenge, but a coordination and data infrastructure challenge. There are four key actions for the UK government to take:

- The DfT, UK Hydrographic Office and Maritime and Coastguard Agency should collaborate to lay the groundwork for machine-readable data availability by working with Community System Providers to agree and publish a minimum UK data specification for predictive berth readiness at ports based on international standards (e.g. DCSA Port Call 2.0[29]).
- DfT should work with industry and regional transport bodies to explore defining a standard (or extending an existing one) for interoperable data exchange around node congestion for road freight (such as depots, ports, warehouses, terminals and charging locations). It should ensure this is interoperable with the equivalent port and rail standards.
- Define a phased mandate for adoption of these data standards, primarily focused on major ports and their associated freight corridors.
- The Competition and Markets Authority could also address operator fears that sharing operational data may be interpreted as collusion by issuing formal guidance on categories of data exchange (and associated conditions).

3. Support aggregation to enable optimisation to scale across organisations

Unlocking the ability of small and medium-sized operators to leverage AI optimisation requires mechanisms that allow optimisation to operate across organisational boundaries. Without such mechanisms, further improvements in route optimisation alone are unlikely to deliver proportional reductions in emissions. Government should:

- Design policy to support aggregation mechanisms that allow small operators to participate in co-optimisation (e.g. shared networks, sector-specific aggregators or intermediated coordination) while avoiding over-centralised platforms that undermine resilience or competition.
- Stimulate new aggregation services via innovation competitions and cross-modal demonstrators, using these to encourage Transport Management System providers to support open APIs and shared signals.
- Direct small grants towards onboarding and integration costs with aggregation mechanisms like pallet networks, which SMEs cite as a primary blocker.

Appendices

These appendices highlight relevant considerations and emerging themes identified during the evidence review that fall outside the scope of the main analysis.

Common data sources

A wide range of data sources and digital platforms already underpin optimisation in UK freight operations across road and maritime modes.

For road freight, commonly used data sources include vehicle telematics (location, speed, fuel use), routing and dispatch data, delivery schedules, and driver behaviour data. These are often supplemented by public or quasi-public contextual datasets, such as real-time and historical traffic flows (e.g. national highways data), incidents and roadworks, weather forecasts, low-emission zone boundaries, and vehicle restrictions. In research and technology development, standardised benchmark datasets, such as the Solomon Vehicle Routing Problem with Time Windows (VRPTW) instances, are widely used to test and compare optimisation algorithms, even though they are simplified representations of real-world conditions.

For maritime freight, data typically includes vessel tracking data (e.g. AIS), port call records, berth availability, cargo handling schedules, and fuel consumption profiles. These are increasingly combined with external data such as weather and sea state, tidal information, port congestion indicators, and emissions reporting data to support route planning, speed optimisation, berth allocation, and port-wide scheduling.

Related technologies

Alongside raw data, a growing number of digital platforms and sector-specific initiatives play an enabling role. These include:

- **Commercial transport management systems, fleet management platforms, and port community systems**, which aggregate operational data and embed optimisation tools directly into day-to-day planning.
- **Sector led initiatives and research programmes**, such as the Smart Energy Data Service (SENSE)[30], aim to make EV and energy use data more visible and actionable across freight and logistics.

Several applications of AI and ML are frequently cited as promising, but fall outside the near-term optimisation opportunities assessed in this pack:

- **Predictive and autonomous maintenance**, aimed at reducing downtime and improving asset utilisation rather than directly reducing freight movements.

- **Smart charging and vehicle to everything applications**, which may become more important as freight electrification increases pressure on depot and grid capacity.
- **Traffic system optimisation**, linking freight planning with wider transport system control.

At present, these applications are best understood as enablers or complements to optimisation.

Design risks and trade-offs

The evidence also highlights several risks that cut across all optimisation approaches:

- **Skills and capability gaps:** Many organisations lack the capacity to deploy and interpret optimisation tools. Targeted, role-specific training focused on operational use in the freight sector may be more effective than supporting broad AI upskilling.
- **Security and systemic exposure**: Greater reliance on digital optimisation, AI, and digital twins increases the potential impact of cyber-attacks or system failures, particularly as coordination across actors deepens.
- **Efficiency versus resilience**: systems that are highly optimised for efficiency can become less robust to disruption if resilience is not built in – e.g. developing fallback mechanisms for when systems fail and being adaptable to sudden change in circumstances or constraints. Operators face trade-offs between maximising efficiency and maintaining flexibility, redundancy, and recoverability.

Additional detail on types of optimisation

Heuristics, rules-based and spreadsheet-driven approaches (non-AI):

"Use fixed rules and human judgement based on experience"

- Many freight operators, particularly small and medium-sized fleets, still rely on manual or semi-automated approaches. These include fixed rules (e.g. driver familiarity, standard delivery sequences), spreadsheet-based planning, and simple decision support tools. These approaches reflect operational experience rather than formal optimisation and remain widespread across the sector.
- These approaches are often described as heuristics - methods that use rules of thumb or simple approaches to find workable solutions quickly, rather than optimal ones. They are designed to cope well with constraints such as limited data, time pressure, or changing conditions.
- Rules-based and heuristic approaches are typically low-cost, flexible, and easy to implement, making them attractive entry points for digital optimisation. However, they miss lots of opportunities for efficiency, do not scale well as problems grow more complex, and struggle to handle uncertainty or coordination across many vehicles or operators.

Exact optimisation methods/mathematical programming (e.g. mixed-integer linear programming):

"Create a mathematical model of each problem and find the provably correct solution"

- Exact optimisation methods use mathematical formulations to identify the best possible solution to a well-defined problem, subject to explicit constraints (such as vehicle capacity or delivery time windows). In freight contexts these are widely used for routing, scheduling, and load allocation.
- For small and medium-sized problems, exact methods are highly reliable, transparent, and interpretable, and often deliver optimal solutions. They form the backbone of many commercial optimisation tools in use today. However, they struggle to scale to larger problems, and the simplifications of the problem required to make it possible to find an exact mathematical solution sometimes mean that the calculated solution does not work as intended in the real world.

Evolutionary algorithms (e.g. metaheuristics**,** genetic algorithms):

"For each problem, iteratively try lots of solutions, combining good ones and discarding bad ones"

- Evolutionary algorithms represent a more advanced form of heuristic search. Rather than applying fixed rules, they use mechanisms inspired by evolution to iteratively explore the potential solution space.
- Within freight, evolutionary algorithms are often used to intelligently search for high quality solutions to complex optimisation problems where exact mathematical methods are too slow or rigid.

Reinforcement learning and policy learning

"Learn effective decision strategies through feedback over time rather than solving each problem from scratch"

- Reinforcement learning (RL) focuses on learning decision-making strategies through receiving feedback from interaction with an environment, often over many simulated scenarios. It is particularly suited to sequential and dynamic problems, where decisions taken now affect future outcomes.
- In freight, RL has been explored for large-scale, highly dynamic problems in simulation and digital twin environments, but has yet to be widely used at scale.
- While RL has the greatest flexibility (and therefore the greatest potential), that also means it is the hardest to set up robustly, which has limited its application in freight so far.

Machine learning (ML) and deep learning (e.g. prediction and approximation):

"Predict the near future from patterns in past data"

- While not specifically a form of optimisation, machine learning methods are commonly used in freight logistics to indirectly support optimisation by improving forecasts and estimates that feed into planning decisions. Typical applications include predicting demand, estimating travel or arrival times, and forecasting congestion or delays. Their value lies in improving inputs and reducing uncertainty, not in directly deciding routes or schedules.
- Deep learning is a subset of ML that is particularly effective where data is large-scale or complex (e.g. high-resolution traffic or sensor data), but it serves a similar supporting role.

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